

IRVIN CHADRAA

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EDUCATION

New York University - Courant Institute of Mathematical Sciences

Master of Science in Mathematics - GPA: 3.89

May 2025

Bachelor of Science in Mathematics & Minor in Computer Science - GPA: 3.63

May 2023

- Founder's Day Award, University Honors Scholar, Dean's List (All Semesters)

Relevant Coursework: Machine Learning, Deep Reinforcement Learning, Computer Vision, Statistics, Stochastic Analysis

TECHNICAL SKILLS

Programming: Python, MATLAB, C++, SQL, Scala

Machine Learning: PyTorch, CUDA, TensorFlow, Scikit-Learn, OpenCV

Tools & Infrastructure: Linux/Bash, Git, SLURM, Spark, Docker

EXPERIENCE

NYU Computational Intelligence, Learning, Vision, and Robotics Lab

June 2025 - Present

AI Research Engineer

- Developing and evaluating predictive models and control policies in PyTorch for long-horizon decision-making in autonomous systems, using visual encoders and world models to forecast system behavior.
- Training diffusion policies on latent-space embeddings to improve sample efficiency and runtime for model-predictive planning in navigation tasks, experienced with end-to-end video generation model development.
- Building and running large-scale model-training pipelines on multi-GPU HPC platforms using SLURM, Docker, and Bash scripting in Linux environment.

NYU Tandon School of Engineering

Sep 2023 - Dec 2025

Graduate Researcher

- Led a 15-person graduate research team applying reinforcement learning to autonomous trading decisions, engineering alpha signals via PCA and discrete wavelet transform (DWT) preprocessing on time-series market data.
- Designed Bayesian hyperparameter calibration framework in PyTorch and sliding-window cross-validation scheme to evaluate model robustness under real-world, non-stationary conditions, yielding 28% increase in net profit.
- Benchmarked model performance against ARIMA and PCA-only baselines through systematic backtesting on historical market data, improving cumulative returns by 15% and 5.6% respectively.

NYU Center for Urban Science and Progress

June 2023 - May 2025

Graduate Researcher - funded by Office of Naval Research

- Developed a game-theoretic optimal transport algorithm in Python and MATLAB for selectively registering a subset of features in high-dimensional remote sensing data, including images and LiDAR point clouds.
- Published two first-author papers (*ISPRS Annals*) and presented results at *20th 3D GeoInfo Conference* in Japan.
- Experienced with flow-based generative models and high-dimensional convex/non-convex optimization.

Isomer Space Program

Feb 2022 - June 2023

Flight Software Engineer

- Co-founded a student-led nonprofit to design and build a CubeSat that detects cosmic radiation in low-Earth orbit, raising over \$30k in crowdsourced funds and delivering the satellite to SpaceX for integration testing.
- Built flight software in CircuitPython including task scheduler, power/boot state management, I2C sensor and payload drivers, and a UHF radio driver for telemetry and command uplink.
- Implemented safety-critical fault handling and low-battery/burn-wire logic, then validated system behavior through hardware-in-the-loop (HITL) testing.

iRobot

Jun 2021 - Dec 2021

Machine Learning Intern

- Finetuned object detection model for Roomba j7+ on custom datasets with TensorFlow, achieving 20% accuracy gains on obstacle detection and avoidance benchmarks, contributing to success of iRobot's top-selling product.
- Experienced in Agile development environment; implemented computer vision pipelines using OpenCV for ground truth data annotation and feature extraction.

PUBLICATIONS

Partial Optimal Transport for Co-Registration of Partially-Overlapped Point Clouds

- Chadraa, I., Tabak, E., & Laefer, D.F. (2025)
- Published in *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences 2025*.

Initial Bounding of Partial Optimal Transport for Point Cloud Co-Registration

- Chadraa, I., Zhu, X., Tabak, E., & Laefer, D.F. (2026)
- Published in *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences 2026*.

TEACHING EXPERIENCE

NYU Tandon School of Engineering

Fall 2024 - Spring 2025

Graduate Teaching Assistant

- VIP-GY 5000: ML and Time-Series Forecasting for Active Portfolio Management

NYU Courant Institute of Mathematical Sciences

Fall 2023 - Fall 2024

Graduate Teaching Assistant

- MA-UY 4414: Applied Partial Differential Equations (Fall 2023)
- MA-UY 4434: Applied Complex Variables (Spring 2024)
- MA-UY 4444: Intro to Math Modeling (Fall 2023, Fall 2024)

Boston College Morrissey School of Arts and Science

Fall 2020

Math Department Grader

- MATH-1100: Calculus I
- MATH-1101: Calculus II

PROJECTS

LLMs with Quaternion Networks — *PyTorch, BERT, Transformers, HuggingFace*

- Researched and developed quaternion-augmented transformer models for sentiment analysis and zero-shot time series forecasting, achieving 75% gain in memory efficiency over traditional BERT and Informer models.

RAG-Based Chatbot on Financial Data — *PyTorch, LangChain, FAISS, LLaMa-2-13B*

- Engineered a RAG-based, quantized LLM chatbot using embedding models and semantic search across structured and unstructured data (stock prices, SEC filings, Glassdoor reviews) to generate insights on S&P500 companies.

Quality of Life in NYC Boroughs — *Scala, Apache Spark, Apache Hadoop, Tableau*

- Used Scala in Apache Spark client shell on Hadoop (HDFS) to build ETL pipelines analyzing large-scale datasets on property value, EMS dispatch frequency, and water quality, then presented visual findings with Tableau.